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RESEARCH

The Effectiveness of Video Modeling on Teaching Physically Active Cooperative Play to Children with Autism Spectrum Disorder

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Abstract
The purpose of this study was to investigate the effectiveness of a video modeling–based intervention package for teaching physically active cooperative playground games to preschool children diagnosed with autism spectrum disorder (ASD). Specifically, this study examined whether the intervention package improved the game performance of three preschool-aged children with ASD. A multiple probe design across participants was employed to assess the effects of the intervention on two traditional playground singing games (i.e., Kutu Kutu Pense and Benim Bir Atım Var). Participants were three preschool children with ASD (two boys and one girl), aged 4–5 years, recruited from an inclusive preschool. The dependent variable was the number of correct task-analysis steps performed during game play while the song/chanted rhyme was ongoing. The independent variable was a teacher-created video modeling–based intervention package that featured age-matched peer models and included visual supports, prompting, and reinforcement. Results indicated that all three children increased the number of correct steps for each game after the intervention package was introduced. Furthermore, during the maintenance phase, each child maintained performance, and generalization probes suggested transfer of the learned routines to different settings and peers. Social validity data from children, parents, and teachers indicated positive ratings of the goals, procedures, and outcomes of the intervention.

Keywords Autism Spectrum Disorder · Video modeling · Cooperative play · Preschool

Recent surveillance estimates from the Centers for Disease Control and Prevention (CDC, 2025) indicate that approximately 1 in 31 children aged 8 years has been identified with autism spectrum disorder (ASD). ASD is a neurodevelopmental dis-

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order characterized by limitations in social communication and interaction, along with restricted interests and repetitive behaviors (American Psychiatric Association [APA], 2013). Variability in these core features also results in differences in the level of support required by individuals diagnosed with ASD (APA, 2013).

Play is a central developmental context in early childhood. It supports social-emotional, physical, motor, language, and cognitive development and provides an important foundation for later learning and participation (Fromberg & Bergen, 2015; Hart Barnett, 2018; Movahedazarhouligh, 2018). In typically developing children, play skills often progress along a predictable social play hierarchy, such as solitary, parallel, associative, and cooperative play (Howes & Matheson, 1992). However, this progression frequently differs for children with ASD, as the core characteristics of ASD may interfere with the natural development of play skills (Barton, 2016; Charlop et al., 2018). Although social interaction and communication are often emphasized as primary areas of need, play represents a critical developmental context in which these challenges become particularly apparent (Barton, 2015; Fedewa et al., 2024). For instance, children with ASD often display considerable differences in both the frequency and quality of their play behaviors compared with typically developing peers (Charlop et al., 2018; Francis et al., 2019). In particular, they may experience difficulties initiating and maintaining interactions with peers during play and other social activities (Gal et al., 2016; Shire et al., 2020). These challenges are commonly associated with communication difficulties, differences in social relationships, and restricted interests, which may limit opportunities for shared play and sustained peer engagement (García-Gómez et al., 2020; Jung & Sainato, 2015; Watkins et al., 2017). Such barriers to interaction can contribute to social exclusion during both preschool and later childhood (Dean et al., 2014; Ulke-Kurkcuoglu, 2015). Given the developmental importance of play, it is critical for children with ASD to acquire skills that support participation in increasingly complex play contexts and social interactions at an early age (Barton & Wolery, 2008; Del Giudice, 2014; Fedewa et al., 2025).

Researchers have emphasized that play skills in children with ASD often require systematic instruction (Boutot et al., 2005; Jung & Sainato, 2013; Quigley et al., 2018). Prior research has examined a range of instructional targets, including symbolic play (Barton & Wolery, 2010; Barton & Pavilanis, 2012; Kasari et al., 2006; Lee et al., 2019), functional and sociodramatic play (Barton & Wolery, 2008; Goldstein & Cisar, 1992), pretend or role play (MacDonald et al., 2009; Reagon et al., 2006), and manipulative play (Groskreutz et al., 2011). Although these play types are well documented, research on teaching physically active cooperative play to preschoolers with ASD in inclusive school settings remains limited (Fedewa et al., 2025; O'Guinn et al., 2025).

Video-Based Interventions

Technology-based instruction has become increasingly prominent in ASD research and practice (Çınar & Sazak Duman, 2025). Advances in digital tools have broadened the scope of video-based instruction to include video modeling, video self-modeling, animation-based modeling, and virtual reality (VR)-based interventions.

Because information and communication technologies (ICT) are widely available in educational settings, video modeling can be implemented in a cost-effective manner (Kocaoz et al., 2019). Furthermore, it allows teachers to adapt content easily to meet individualized instructional goals.

Grounded in Bandura's Social Learning Theory (Bandura, 1977), video modeling supports observational learning by presenting target behaviors in a structured, consistent, and repeatable format, thereby increasing opportunities for repeated exposure. Video modeling is an evidence-based practice for teaching a broad range of skills to individuals with ASD (Bellini & Akullian, 2007; Nikopoulos & Keenan, 2003; Shukla-Mehta et al., 2010; Steinbrenner et al., 2020). Prior studies have reported positive outcomes for daily living skills (Hong et al., 2015), communication initiation (Kabashi & Kaczmarek, 2017; Maione & Mirenda, 2006), play behaviors (Charlop-Christy et al., 2000; Çınar & Sazak Duman, 2025; Kurnaz & Yanardag, 2018; Reagon et al., 2006; Sani-Bozkurt & Ozen, 2015), and social skills (Kocaoz et al., 2019). However, more studies are needed on teacher-delivered, low-cost video modeling interventions designed to teach culturally embedded, physically active group games to preschoolers with ASD (Petursdottir & Gudmundsdottir, 2023).

Current Study

From an educational technology perspective, mobile devices make it feasible for teachers to create and use brief instructional videos in real school settings. Yet evidence on teacher-implementable video modeling for complex group play routines remains limited (Wolfberg et al., 2015). Teaching these routines may be especially valuable because they reflect naturally occurring playground activities and create structured opportunities for performing coordinated actions with peer. Physically active cooperative play is a traditional children's group play activity in which participants sing or chant a rhyme while performing coordinated actions in a structured sequence. In the present study, the targeted playground games can also be conceptualized as rule-governed cooperative social play routines because they require coordinated actions with peers (e.g., holding hands, synchronized stepping) and responding to shared auditory cues embedded within a song-based script. Therefore, the present study investigated the effectiveness of a teacher-implemented video modeling–based intervention package for teaching two physically active cooperative play routines to preschool children with ASD. For this investigation, the following research questions were investigated.

1) To what extent does a video modeling–based intervention package increase the number of correct steps performed by preschool children with ASD during two traditional playground singing games?
2) Were the goals, procedures, and outcomes of the video modeling–based intervention package rated as socially valid by children, parents, and teachers?

Method

Participants and Settings

The researcher obtained ethical approval from the Institutional Review Board (IRB) of a state university in Türkiye and official permission from the National Education Directorate. The study included three children diagnosed with ASD who attended an inclusive preschool in the central region of Türkiye. The inclusion criteria required participants to demonstrate the following prerequisite skills: (a) following verbal instructions, (b) attending to a video for at least 3 min, (c) maintaining attention to visual and auditory stimuli for at least 3 min, (d) understanding the concepts of left and right, and (e) imitating both fine and gross motor movements. Before the intervention began, informed consent was obtained from the participants' parents. Pseudonyms were used for all participants to protect confidentiality. Demographic information about the participants is presented in Table 1.

Efe was a 5-year-old boy diagnosed with ASD. He had received services at a special education and rehabilitation center since age 1 and had attended an inclusive preschool for 2 years. Efe was able to independently complete key daily living skills, including eating meals, dressing and undressing, and toileting. He recognized numbers, matched numbers with objects from 1 to 5, and identified primary colors and geometric shapes. In terms of fine motor skills, he was able to cut randomly with scissors and draw lines while holding a pencil properly. In terms of gross motor skills, he could ride a tricycle and run in a coordinated manner. Efe was able to imitate movements demonstrated by an adult or observed in a video, and he enjoyed playing ball games.

İkra was a 5-year-old girl diagnosed with ASD. She had received services at a special education and rehabilitation center since age 3 and had attended an inclusive preschool for 2 years. İkra demonstrated both receptive and expressive language skills and was able to provide logical answers to questions addressed to her. She independently performed the expected self-care skills for her age group. She recognized numbers, counted rhythmically up to 20, and identified primary colors, including red, blue, yellow, and green, as well as geometric shapes, including square, triangle, circle, and rectangle. İkra enjoyed pretend play activities, such as playing house.

Sinan was a 4-year-old boy diagnosed with ASD. He had received services at a special education and rehabilitation center since age 1 and had attended an inclusive preschool for 1 year. Although Sinan had fluent speech, most of his words were difficult to understand. He was able to attend to the person addressing him, follow instructions, and carry them out appropriately. He recognized numbers up to 20 and matched numbers with objects from 1 to 10. He also identified primary and second-

Table 1 Demographic information of participants

Name	Gender	Ages	Primary Exceptionality	Length of school attendance
Efe	Male	5	ASD	2 years
İkra	Female	5	ASD	2 years
Sinan	Male	4	ASD	1 years

ary colors, geometric shapes, and three-dimensional objects. His fine motor skills included cutting paper with scissors, scribbling with a pencil, and threading beads. His gross motor skills included climbing stairs, riding a tricycle, throwing a ball, and catching a ball. Sinan enjoyed playing alone with toys as well as engaging in physically active games.

Setting

The study was conducted in an inclusive preschool located in the central region of Türkiye. Sessions took place in two designated areas within the preschool. Video-viewing sessions were conducted in an individual education classroom measuring 3×2 m. The room contained a small table and two chairs, which provided a focused environment for viewing the videos. Instructional and game-playing sessions were conducted in a multipurpose hall measuring 6×11 m. The hall included soft tatami mats that allowed for safe and appropriate movement during the games. It also contained a large mirror and a television mounted above the mirror, which facilitated the visual presentation of the video models and allowed participants to observe themselves during practice. The minimal furniture in the smaller room and the visibility of the adjacent kitchen from the hall did not interfere with participants' attention to the instructional stimuli or with the implementation of the instructional protocol.

Selection of Target Games

Targeted games included traditional Turkish singing games such as *Kutu Kutu Pense* (similar to *Ring a Ring o' Roses*) and *Benim Bir Atım Var* (an action song similar to *The Hokey Pokey*). These games involve a structured sequence of movements, including synchronized stepping, turning, and imitation of rhythmic actions. Movements are timed to a shared song- or chant-based script. Although the games are typically accompanied by singing or chanting, the primary measure in this study was participants' accurate completion of the game steps during the song, rather than vocal production of the lyrics.

These games were selected in consultation with special education teachers, preschool teachers, and parents. Also, the selected games aligned with criteria commonly emphasized in the literature, including comparable difficulty, physical activity, anticipated enjoyment, and feasibility for implementation in the school setting (Boutot et al., 2005; Wolfberg, 2003). Thus, the games were considered feasible to implement and developmentally appropriate for each participant based on parent and classroom teacher reports.

Task Analyses

The first author developed separate task analyses for Game A and Game B. An expert special education teacher with 10 years of experience and a special education faculty member reviewed the task analyses for accuracy and completeness. The task analyses specified the sequential and cooperative steps required to complete each game. The task analysis for Kutu Kutu Pense consisted of 12 steps, whereas the task analysis

Table 2 Task analysis of Kutu Kutu Pense (Game A)

Steps	Kutu Kutu Pense Task Analysis
1	Holds the hands of the friends standing on both sides using both hands.
2	Steps to the right and walks sideways.
3	Stops in place.
4	When they hear their own name in the song, releases their friends' hands.
5	Turns around.
6	Holds the hands of the friends standing on both sides using both hands.
7	Steps to the left and walks sideways.
8	Stops in place.
9	When they hear their own name in the song, releases their friends' hands.
10	Turns to face forward.
11	Holds the hand of the friend standing next to them.
12	Walks sideways by stepping to the side until the song ends.

Table 3 Task analysis of Benim Bir Atım Var (Game B)

Steps	Benim Bir Atım Var Task Analysis
1	Holds the hands of the friends standing on both sides using both hands.
2	Squats down.
3	Stands up.
4	Extends arms forward and steps forward.
5	Moves arms backward and steps backward.
6	Releases their friends' hands.
7	Crosses arms (folds arms).
8	Extends arms out to the sides.
9	Claps.
10	Lowers arms to the sides.
11	Stomps feet on the ground.
12	Jumps in place.
13	Stands still.
14	Crosses arms (folds arms).

for Benim Bir Atım Var consisted of 14 steps. The full task analyses for each game are presented in Tables 2 and 3. The task analyses were included as part of the video modeling–based intervention package and served multiple instructional and measurement purposes. Specifically, they were used to structure the video models, develop the visual cards, guide step-by-step instruction, and record participants' performance.

Video Model Development

The researchers developed the video models following evidence-based procedures (Cox & AFIRM Team, 2018). These procedures were drawn from Autism Focused Intervention Resources and Modules (AFIRM), which provide systematic, step-by-step guidance for planning, implementing, and monitoring evidence-based practices.

Once the target play skills and task analyses were finalized, the researchers used them as instructional scripts to guide the creation of peer model videos. The researchers identified three typically developing peers, one girl and two boys, to serve as models and obtained informed consent from their parents before the recording process. The researchers selected peers who matched the participants in terms of gender, age, and physical characteristics to enhance perceived similarity and potential learning.

The researchers used a Samsung device with a 50-MP camera capable of 4 K resolution, 2160p at 30 fps. An ASUS laptop with a 15.6-inch screen served as the delivery platform to optimize visual and auditory clarity. The videos were recorded in the multipurpose hall of a special education kindergarten, using a plain white background and an area cleared of distracting stimuli. During the recording sessions, the peer models performed the games step by step, allowing the researchers to capture screenshots of each task step for use as supplementary visual aids. The final Kutu Kutu Pense video, which lasted 1 min 50 s, required three recording sessions, whereas the final Benim Bir Atım Var video, which lasted 2 min, required two recording sessions. Finally, two special education experts and one instructional technology expert reviewed the finalized videos using a modified video panel checklist (Kocaoz et al., 2019). Their positive feedback confirmed the suitability of the videos for the intervention.

Visual Cards

The first author created visual cards measuring 16 × 9 cm for use during step-by-step instruction and discussion of the game routines. One card was developed for each task-analysis step using screenshots from the video models and a Microsoft PowerPoint slide template. The cards were laminated to ensure durability. A classroom teacher and a special education faculty member subsequently reviewed the visual cards and verified their accuracy and appropriateness.

Dependent Variable and Measurement

The dependent variables of this study were participants' performance in completing the behavior chains required for two traditional children's games: Kutu Kutu Pense (Game A) and Benim Bir Atım Var (Game B). Participants' performance was measured as the number of correctly performed steps within the defined task analysis for each game.

Independent Variable

The independent variable was a multicomponent video modeling–based intervention package designed to teach physically active cooperative play routines. This intervention integrated peer-model video modeling with explicit instruction for each component of the target play skills. To further support the learning process, the researchers used visual cards depicting each task-analyzed step of the games.

Data Collection Tools

The researchers used three data-collection instruments to evaluate the intervention. First, they developed game-specific data collection record forms for Kutu Kutu Pense and Benim Bir Atım Var. These forms were derived directly from the task analyses and were used to measure the dependent variable. The researchers used the forms across all experimental phases to record the number of steps performed correctly. Second, social validity data were collected through questionnaires designed for children, parents, and teachers. For this purpose, the first author developed three questionnaires after the completion of the study, drawing on prior video modeling research (Bellini & Akullian, 2007; Kocaoz et al., 2019). The teacher form comprised six closed-ended and two open-ended questions, the parent form included seven closed-ended and one open-ended question, and the child form included eight items in a simplified yes/no format. Finally, the researchers used procedural fidelity checklists to monitor the consistency of intervention delivery. These checklists were based on the specific steps of the instructional protocol and allowed the researchers to verify that all sessions were implemented as planned and to calculate procedural fidelity percentages.

Implementer and Observers

The implementer held bachelor's and master's degrees in special education and had 21 years of teaching experience. The implementer's primary responsibilities were to: (a) design the study and identify the target skills in collaboration with the research team; (b) prepare lesson plans and instructional materials used during the intervention; (c) develop the video models and accompanying supports, such as examples and visual cards; (d) implement each study phase, including baseline, intervention, maintenance, and generalization, as scheduled; (e) collect participant performance data and procedural fidelity data; (f) participate in observer training; and (g) consult with other research team members as needed.

The observers were a school counselor with 12 years of experience and a preschool teacher with 15 years of experience. Both were employed at the school where the study was conducted. The first author trained the observers on the study's measurement system, including the operational definitions of correct and incorrect steps for each game and the scoring procedures for the data-recording forms. The first author also reviewed the procedural fidelity checklists with the observers and provided practice opportunities using sample video-recorded sessions until they demonstrated consistent scoring. During the study, the observers independently scored 30% of sessions in each phase to calculate interobserver agreement and procedural fidelity.

Experimental Design

This study employed a multiple-probe design across participants to examine the effectiveness of a video modeling–based intervention package in teaching physically active cooperative play routines to children with ASD. During the initial design phase, the researchers incorporated quality indicators for single-case research to strengthen methodological rigor (Horner et al., 2005). This design is appropriate for

demonstrating a functional relation between the intervention and the targeted skills (Gast et al., 2014). In addition, the multiple-probe format reduces the need for continuous measurement by allowing intermittent probe sessions across time while still documenting changes in performance (Gast, 2014; Tekin-İftar, 2018).

Procedure

Baseline

Data collection began with the baseline phase. During baseline probe sessions, the researchers used the single-opportunity method to assess participants' ability to perform the entire task-analysis chain. In this method, the researcher delivered one instruction and allowed a 3-s response interval for the participant to respond. The session was terminated immediately when the participant responded incorrectly at any step. The incorrect step and all subsequent steps were recorded as incorrect (–) on the data collection form. The researchers conducted baseline sessions once per day, with one trial per session. Data collection continued until stable baseline performance was demonstrated for each target game, defined as a minimum of three data points with no accelerating or decelerating trend.

Intervention

Instruction for Game A, Kutu Kutu Pense, began after stable baseline data were obtained. The video modeling–based intervention package was implemented through a structured instructional sequence. At the beginning of each session, the implementer arranged the instructional materials, including the laptop, video model, visual cards, data collection form, and reinforcer. The implementer then gained the child's attention, introduced the session goal, and explained that the child would watch a video of the target game and then practice the game with peers. The implementer also explained the reinforcement contingency before instruction began.

The child first watched the complete video model on the laptop without interruption. If the child's attention shifted away from the screen, the implementer redirected the child's attention to the video. The video was then presented a second time, and the implementer paused it at each task-analysis step. For each step, the corresponding visual card was shown, the expected action was briefly described, and the child's attending responses were reinforced. If the child did not attend to the video or visual card, the implementer repeated the relevant video segment and presented the visual card again.

After the video-based instruction, the child was invited to practice the target game with peers in the multipurpose hall. During the intervention, the implementer used the task analysis to guide step-by-step instruction and determine whether the participant completed each step correctly. Kutu Kutu Pense included 12 task-analysis steps, whereas Benim Bir Atım Var included 14 task-analysis steps.

During practice, the song or chant was initiated, and the child was given a 3-s response interval for each task-analysis step. Correctly completed steps were recorded as correct on the data collection form. If the child did not respond or performed a step

incorrectly, the implementer provided a verbal prompt to support completion of the step. Correct responses and participation were followed by verbal praise and natural reinforcement embedded in the play activity. Verbal prompts were faded across instructional sessions as the child demonstrated increased independence. At the end of each session, the child received the identified reinforcer. One instructional session was conducted per day, and each session included one trial. The same instructional procedures were used for Game B, Benim Bir Atım Var.

Maintenance

The researchers collected maintenance data to assess whether the three participants maintained the acquired play skills. Maintenance probes were conducted 7, 14, and 21 days after the final instructional session. The procedures used during maintenance sessions followed the same single-opportunity method used in baseline probe sessions. The results of the maintenance probes were displayed graphically alongside baseline and intervention data to support a comprehensive visual analysis of skill maintenance.

Generalization

The researchers assessed generalization by conducting three probes at different intervals after the maintenance sessions. During these probes, participants performed the games under combined generalization conditions: a different setting, different peers, and a changed group composition. Specifically, the probes were conducted on the school playground with typically developing peers who had not served as video models, and all three participants played together. These conditions allowed the researchers to examine whether participants transferred the acquired cooperative play skills across settings, people, and group composition. Generalization data were collected using the same single-opportunity method used during baseline and maintenance probes.

Data Analysis

Effectiveness data were analyzed using visual analysis. Visual analyses were conducted by examining changes in level, trend, nonoverlap, and variability across phases (Gast & Spriggs, 2010). Data for each phase were reported session by session and displayed in simple line graphs. Each participant's performance was graphically displayed in a multiple-probe design format, with the horizontal axis representing sessions and the vertical axis representing the number of correct responses for each game (Tekin-İftar, 2018). In addition, the researchers calculated the percentage of nonoverlapping data (PND) for each participant as an effect size indicator by comparing baseline and intervention data. Following the analysis of effectiveness data, social validity findings were examined based on responses from three groups: children, parents, and teachers. Responses to closed-ended questions were analyzed separately for each group and summarized descriptively.

Procedural Fidelity and Interobserver Agreement

Procedural fidelity was assessed across all phases to verify that the intervention was implemented as planned. Using phase-specific procedural fidelity checklists, the researchers calculated procedural fidelity by dividing the number of correctly implemented steps by the total number of planned steps and multiplying the result by 100. Procedural fidelity was 100% for all three participants across the assessed sessions. Interobserver agreement (IOA) was calculated using the point-by-point agreement method. The number of agreements was divided by the total number of agreements plus disagreements, and the result was multiplied by 100. IOA was 100% across all assessed sessions for each participant.

Result

Game A: Kutu Kutu Pense

Figure 1 presents the number of correct steps completed by each participant across the baseline, intervention, and maintenance phases for the Kutu Kutu Pense game.

Efe produced 0 correct responses across three baseline sessions. Following baseline, the intervention was introduced, and a total of 10 instructional sessions were conducted. Efe showed somewhat variable performance during the first four sessions, with scores ranging from 5 to 8 correct steps. However, his number of correct steps gradually increased beginning with the fourth session. The last five intervention data points showed a stable pattern, with Efe performing all 12 steps correctly. Efe's mean number of correct steps during intervention was 9.7, with a range of 5 to 12. This data pattern indicates a clear increase in performance level compared with baseline. Furthermore, Efe maintained his performance across three maintenance sessions, completing all steps correctly in each session.

For İkra, baseline performance was 0 correct responses across all baseline sessions. Following the introduction of the intervention package, İkra's performance increased, with a clear improving trend in the number of correct steps completed. A total of 10 intervention sessions were conducted for this target skill. İkra's mean number of correct steps during intervention was 10.6, with a range of 8 to 12. The data pattern during the final six intervention sessions was stable, with İkra performing all 12 steps correctly. Following intervention, three maintenance probes were conducted, and İkra completed all steps correctly in each maintenance session.

Similarly, Sinan completed 0 correct steps across three baseline sessions. Following three consecutive stable baseline data points, the video modeling–based intervention package was implemented across 10 sessions. During intervention, Sinan's performance showed a clear upward trend, increasing from low initial levels to criterion-level performance. His mean number of correct steps during intervention was 9.3, with a range of 5 to 12. Performance during the final four intervention sessions was stable, with Sinan performing all 12 steps correctly. After intervention, three maintenance probes were conducted, and Sinan completed all steps correctly in each session.

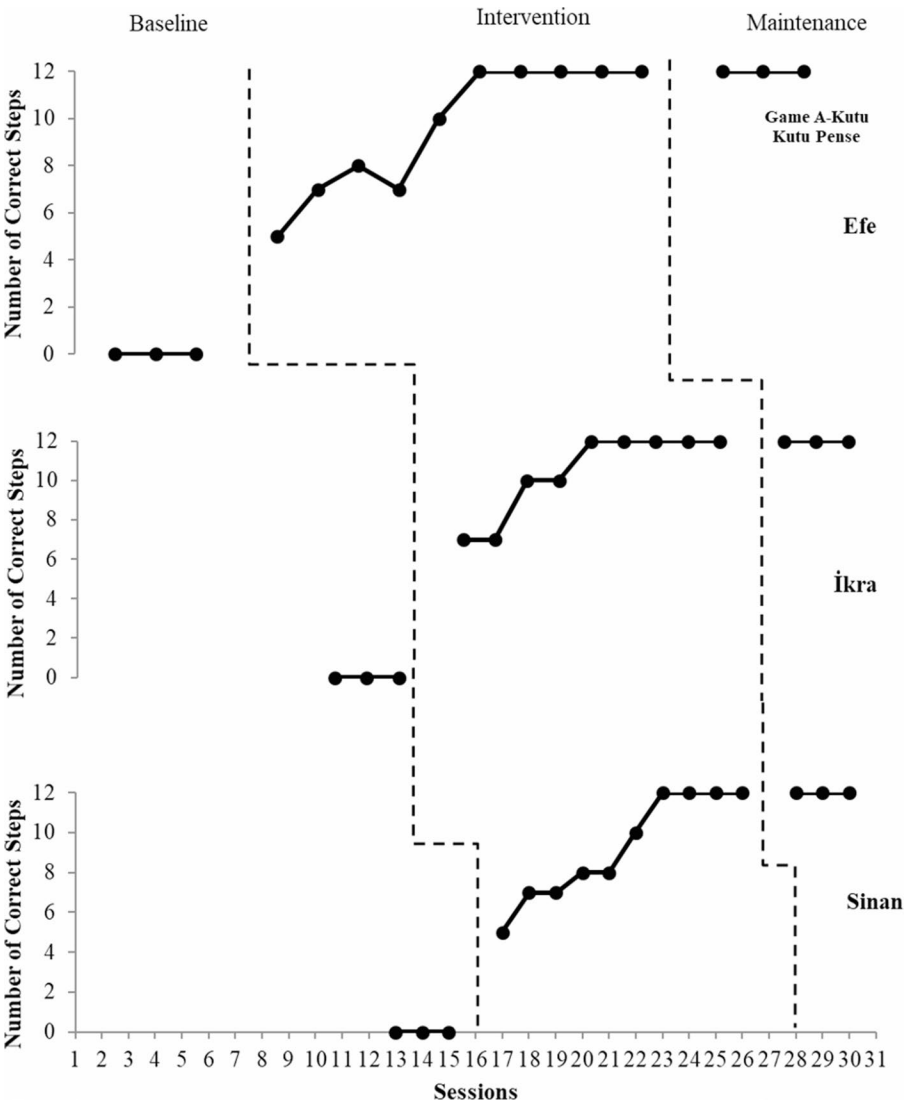


Fig. 1 Number of correct steps completed for the Kutu Kutu Pense game across baseline, intervention, and maintenance phases

Game B: Benim Bir Atım Var

Figure 2 presents the number of correct steps completed by each participant across the baseline, intervention, and maintenance phases for the *Benim Bir Atım Var* game.

Efe completed 0 correct steps across three baseline sessions. Following three consecutive stable baseline data points, the video modeling–based intervention package was implemented across 10 sessions. During intervention, Efe demonstrated a clear increasing trend compared with baseline. His mean number of correct steps during

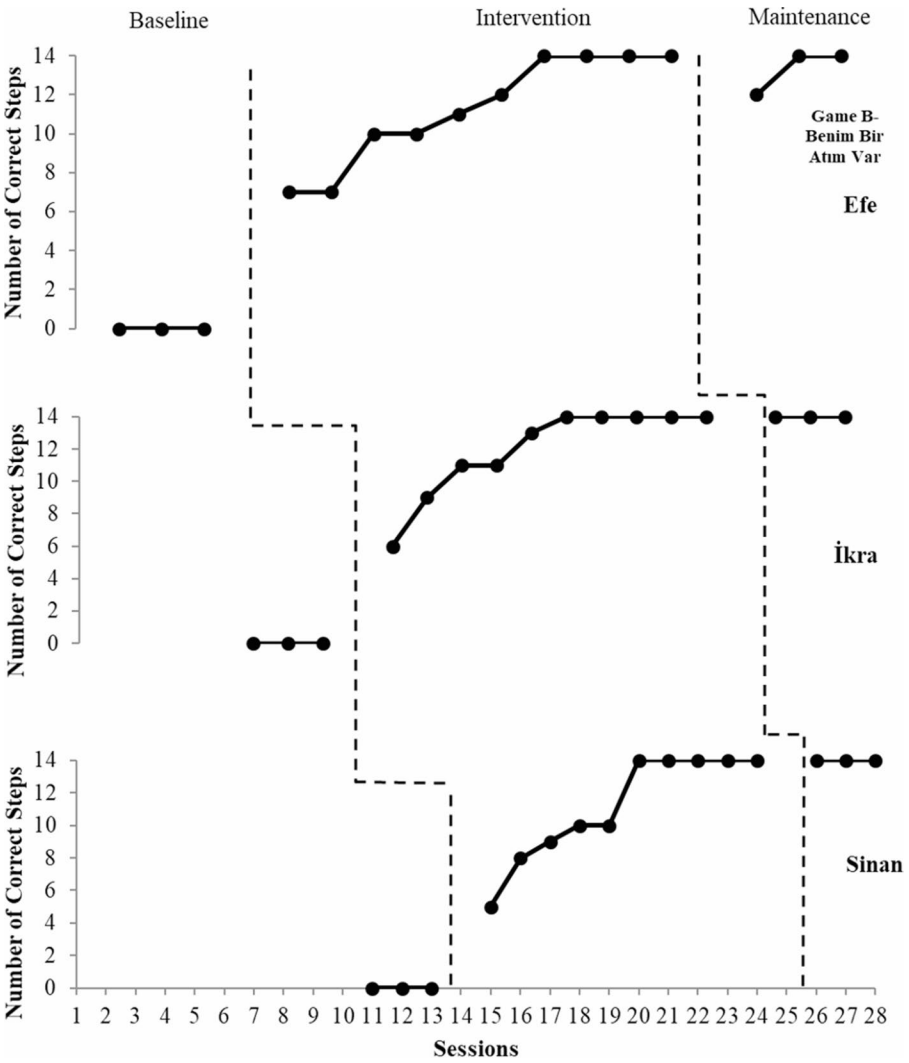


Fig. 2 Number of correct steps completed for the Benim Bir Atım Var game across baseline, intervention, and maintenance phases

intervention was 11.3, with a range of 7 to 14. Performance during the final four intervention sessions was stable at criterion level, with Efe completing all 14 steps correctly. Three maintenance probes were conducted after the intervention, and Efe's mean number of correct steps during maintenance was 13.3.

For İkra, baseline performance was also stable at 0 correct steps across three sessions. Following three consecutive stable baseline data points, the video modeling–based intervention package was implemented across 10 sessions. During intervention, İkra demonstrated steady improvement, increasing from partial completion to criterion-level performance. Her mean number of correct steps during intervention was 12.0, with a range of 6 to 14. Performance during the final five intervention

Table 4 Participant probe-by-probe performance (Number of Correct Steps/Total Steps)

Games	Participant	Probes 1	Probes 2	Probes 3
Kutu Kutu Pense	Efe	12/12	12/12	12/12
	İkra	12/12	12/12	12/12
	Sinan	12/12	12/12	12/12
Benim Bir Atım Var	Efe	14/14	14/14	14/14
	İkra	14/14	14/14	14/14
	Sinan	14/14	14/14	14/14

sessions was stable, with İkra completing all 14 steps correctly. After intervention, three maintenance probes were conducted, and İkra completed all 14 steps correctly in each probe.

Similarly, Sinan completed 0 correct steps across three baseline sessions. Following three consecutive stable baseline data points, the video modeling–based intervention package was implemented across 10 sessions. During intervention, Sinan demonstrated a clear increasing trend, progressing from partial completion to criterion-level performance. His mean number of correct steps during intervention was 11.2, with a range of 5 to 14. Performance during the final five intervention sessions was stable at criterion level, with Sinan completing all 14 steps correctly. After intervention, three maintenance probes were conducted, and Sinan completed all 14 steps correctly in each probe.

Generalization Results

Generalization results, presented in Table 4, indicated that all participants performed both game routines under novel conditions. Across the three generalization probes, each participant completed all task-analysis steps for both games: 12 of 12 steps for Kutu Kutu Pense and 14 of 14 steps for Benim Bir Atım Var. Participants achieved these scores while playing on the school playground, with typically developing peers who had not served as video models, and with all three participants playing together. These findings suggest that participants were able to perform the learned play routines across changes in setting, peers, and group composition.

Social Validity Result

Responses from children, parents, and teachers addressed the acceptability of the instructional methods, the appropriateness of the games, and overall satisfaction with the implementation (Wolf, 1978). Children's reports indicated acceptability of the activities and the video modeling component. All three children reported that they liked participating in the activity, enjoyed both games, Kutu Kutu Pense and Benim Bir Atım Var, wanted to play the games with their friends, enjoyed watching the game videos, and wanted to learn additional games by watching videos.

Parent responses indicated that the targeted games and the video modeling component of the intervention package were acceptable and perceived as beneficial. All three parents reported that they would play the games their child learned at home. They also believed that the physically active cooperative play games contributed to their child's motor development, helped meet daily physical activity needs, and sup-

ported social development. All parents stated that they would like their child to learn additional games and that they would consider using video modeling at home to teach other skills. None of the parents provided additional comments or suggestions about the intervention in the open-ended section.

Teacher responses suggested perceived feasibility of the video modeling component and possible applicability beyond the targeted games. All six teachers reported that they used video modeling in their instructional work with children with ASD and agreed that the traditional games taught in the study supported motor development, daily physical activity, and social development. Teachers also indicated that the games taught were important skills for their students and that students could play these games with peers in other settings. In open-ended responses, teachers noted that watching videos captured students' attention, combining visual and auditory input facilitated learning, and video modeling could be used to teach other domains, including self-care and daily living skills (e.g., handwashing, toothbrushing) and academic skills (e.g., concept instruction). Two teachers suggested shortening the instructional videos and/or reducing the number of steps to prevent student boredom.

Discussion

The findings of this study suggest that a video modeling–based intervention package was associated with improvements in physically active cooperative play routines among three preschool children with ASD. Based on PND calculations, all participants demonstrated 100% PND for both games when baseline and intervention phases were compared, indicating consistent nonoverlap between baseline and intervention data. These findings align with prior video modeling research; however, they should be interpreted as reflecting the effects of a multicomponent intervention package rather than video modeling alone. For example, Boudreau and D'Entremont (2010) demonstrated that video modeling promoted the rapid acquisition of motor actions and verbalizations while supporting generalization to novel settings. Similarly, Dueñas et al. (2019) used joint video modeling in natural classroom environments and reported increases in both scripted and unscripted social play.

Although much of the existing video modeling literature has focused on solitary or pretend play routines, the present study contributes to this literature by targeting physically active cooperative games for preschoolers with ASD. Targeting such games may be particularly relevant because children with ASD are more likely to engage in solitary play even when they are near same-age peers in inclusive environments. Gross motor actions, such as jumping and turning, may make participation more engaging and enjoyable (Veitch et al., 2008). Although this study did not directly measure social interaction, learning culturally familiar playground routines may create additional opportunities for children to participate in everyday peer activities.

Several factors may help explain participants' improvements in play performance. First, participants were highly engaged while watching the videos. This pattern suggests that video viewing itself may have been reinforcing (Delano & Stone, 2014). Second, the use of age-matched peer models may have made the demonstrations

more meaningful and easier to imitate. Third, the researchers filmed the videos in the children's own school environment. This likely reduced the gap between what participants observed during instruction and what they were expected to do during practice. This contextual similarity may have supported participants' engagement during practice. Although this study did not include a component analysis to determine whether the video, prompting, or reinforcement was primarily responsible for behavior change, the instruction was delivered in an explicit and systematic manner. This structured delivery, combined with modeling, prompting, and reinforcement, is consistent with prior research showing that integrated intervention packages can be effective for teaching complex leisure and play skills to individuals with ASD (Yanardağ et al., 2013).

Beyond intervention effects, implementation under routine school conditions is also an important consideration. The use of teacher-created videos supports school-based feasibility by relying on typical classroom resources and aligning with instructor-led implementation conditions. This approach is supported by Palechka and MacDonald (2010), who found that instructor-created video models often led to quicker skill acquisition for some children than commercially produced alternatives. Furthermore, the use of video models within the intervention package may reduce the need for repeated live modeling, which may make the approach feasible for applied school settings.

Limitations and Future Studies

This study has several limitations that should be considered when interpreting the findings. First, the study included three preschool children with ASD in a single school context in Türkiye. This limits the generalizability of the findings to other settings, cultures, age groups, and learner profiles. Future studies should replicate the intervention with larger samples across multiple schools. They should also examine whether the video modeling–based intervention package can be adapted for different cultural contexts, traditional games, and learner profiles.

Second, the intervention was delivered as a multicomponent video modeling–based intervention package that included video modeling, visual supports, prompting, and reinforcement. Because the study did not include a component analysis, the relative contribution of each instructional component cannot be determined. This limits conclusions about the unique contribution of video modeling. Future research should use component analysis or comparative designs to identify the active ingredients of the package.

Third, the target games were accompanied by a song or chanted rhyme. However, the dependent measure focused on accurate completion of game steps and did not assess vocal production of the lyrics or broader social communication outcomes. Therefore, the findings should not be interpreted as evidence of improvement in peer interaction, spontaneous communication, or engagement. Future studies should assess vocal participation and peer interaction as additional dependent variables.

Fourth, generalization probes combined multiple changes at the same time, including setting, peers, and group composition. This approach provided information about performance under more naturalistic conditions. However, it did not isolate which

variable contributed to participants' performance. Future research should test generalization variables separately before combining them in naturalistic probes.

Implications

Play skills represent a critical early developmental context for children with ASD. Many children with ASD experience difficulties engaging in peer-oriented play routines. Therefore, supporting play participation should be considered an important instructional priority in early childhood settings. The findings of the present study provide preliminary support for the use of a teacher-implemented video modeling–based intervention package to teach physically active, rule-governed cooperative game routines.

In this study, the video models featured age-matched peers and were delivered in the school context. This format may help increase attention and motivation during instruction. Video models also offer practical advantages for teachers because they provide accurate demonstrations of the target routine and can be replayed as often as needed until mastery. Therefore, a video modeling–based intervention package may be considered a practical and accessible instructional option for classroom use. However, teachers should recognize that the package included video models together with visual supports, prompting, and reinforcement. Therefore, these components should be planned systematically when implementing the approach in classroom settings. This study also highlights the value of structured implementation procedures. Teachers need to define the target routine clearly and use a task analysis to specify each step. They should also ensure that the model demonstrates the skills correctly and that the setting and materials match the natural play context. In addition, visual cards created from video screenshots can serve as prompts during step-by-step instruction and may help children recall multistep game routines during practice. These supports may be especially useful when the target routine includes many components and requires coordination with peers.

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Declarations

Ethical Approval All procedures involving human participants were conducted in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards. Ethical approval was obtained from the Aksaray University Institutional Review Board (IRB) (Protocol No. 2023-07/72) prior to the initiation of the study, and official permission was also obtained from the Aksaray Provincial Directorate of National Education.

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